# CHANDU PRASAD BHAIRAPU



## AI Developer

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github

## ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in generative AI and workflow automation, from LLM agents and AI-assisted coding with Claude Code to automated workflows that replace manual, multi-step work. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by close work with business stakeholders and a strong foundation in Python development across enterprise and start-up environments.

## EXPERIENCE

### Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

**Kniit GmbH**

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end, automating a multi-step document task: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Delivered as an AI-augmented engineer, using Claude Code as an AI coding assistant with custom skills and agentic coding workflows to generate, debug and test code, versioned in Git and grounded in clean code principles.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) for complex regulatory and technical documents, validated against an SAP-system-derived ground truth; precision/recall raised from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, choosing plain code over LLM calls where it was enough and cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- **Tech stack: Claude Code, LLMs, Python, LangGraph, LiteLLM, MCP, RAG, FastAPI, React, Docker, Azure**

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### GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

**Covestro Deutschland AG**

December 2024 – April 2025 Leverkusen, Germany

- Process mapping with business stakeholders: understood their decision-making workflows, advised on where agentic AI could support them and translated their requirements into the solution approach.
- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference and summarization tasks.
- Developed a React front-end and a Python backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Postman, Docker, GitLab CI/CD, AWS (CDK, EC2, Lambda)**

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### GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

**Covestro Deutschland AG**

May 2024 – November 2024 | Leverkusen, Germany

- Automated generation and retrieval evaluation of a RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK)**

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### Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

**Covestro Deutschland AG**

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

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### Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

**Cognizant Technology Solutions**

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time-series analysis of EV load.
- Developed an LSTM model that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

## PROJECTS

**Project: Order Inbox Agent | LLMs | Google Apps Script | Make | GitHub Actions**

- Automates operational processes run on spreadsheets: order emails, PDFs and Excel files are extracted by an LLM, matched to a product catalog (exact, fuzzy and embeddings) and written to Google Sheets through Google Apps Script; a Make low-code scenario routes low-confidence lines to a person; versioned in GitHub with pull requests, run as scheduled jobs in GitHub Actions with a labelled test set.

## TECHNICAL SKILLSET

- **AI & Automation:** AI coding assistants (Claude Code, custom skills), intelligent agents, LLMs, prompt engineering, LangGraph, LangChain, LiteLLM, MCP
- **Programming, APIs & CI/CD:** Python, FastAPI, REST APIs, React/JavaScript (beginner), Git, GitHub, GitLab CI/CD
- **Cloud & Containers:** Docker, Kubernetes, Azure, AWS (S3, EC2, Lambda, SQS, DynamoDB, CDK)
- **Data & ML:** SQL, Postgres, MongoDB, Neo4j, Pandas, Streamlit, Scikit-Learn, PyTorch
- **Languages:** German (B1), English (Business Fluent), Hindi (Fluent), Telugu (Native)

## EDUCATION & PUBLICATIONS

**Master of Science in Artificial Intelligence** — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

**Bachelor of Technology in Mechanical Engineering (Minor in CSE)** — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

**Publication:** Navigating the Future: An approach of Autonomous Indoor Vehicles